

Day 2: TCP/IP Suite

Task 1 — Wireshark TCP/IP Layers

I opened Wireshark and captured network traffic while refreshing the Flipkart homepage. I checked the Protocol and packet details to identify the TCP/IP layers.

TCP/IP Layer	Protocol / Example	How I Identified It
Application	DNS / HTTP / TLS	It is related to application-level communication
Transport	TCP	TCP header contains source and destination ports
Internet	IP	IP header contains source and destination IP addresses
Network Access	Ethernet / ARP	It contains information related to local network communication

Result:

Wireshark helped me understand how different protocols work at different layers of the TCP/IP model.

Task 2 — OSI vs TCP/IP Model

OSI Model	TCP/IP Model	Real-World Example
Application	Application	WhatsApp message
Presentation	Application	Image/video formatting
Session	Application	WhatsApp/Instagram session
Transport	Transport	TCP
Network	Internet	IP address
Data Link	Network Access	Ethernet / Wi-Fi
Physical	Network Access	Network cable / Wi-Fi signal

Short Explanation

The **OSI model** has **7 layers**, while the **TCP/IP model** has **4 main layers**. In TCP/IP, the Application layer combines the functions of the OSI Application, Presentation and Session layers. The Network Access layer covers functions related to the OSI Data Link and Physical layers.

Task 3 — TCP Segment Header

Note: This task is based on the TCP header screenshot provided by the trainer.

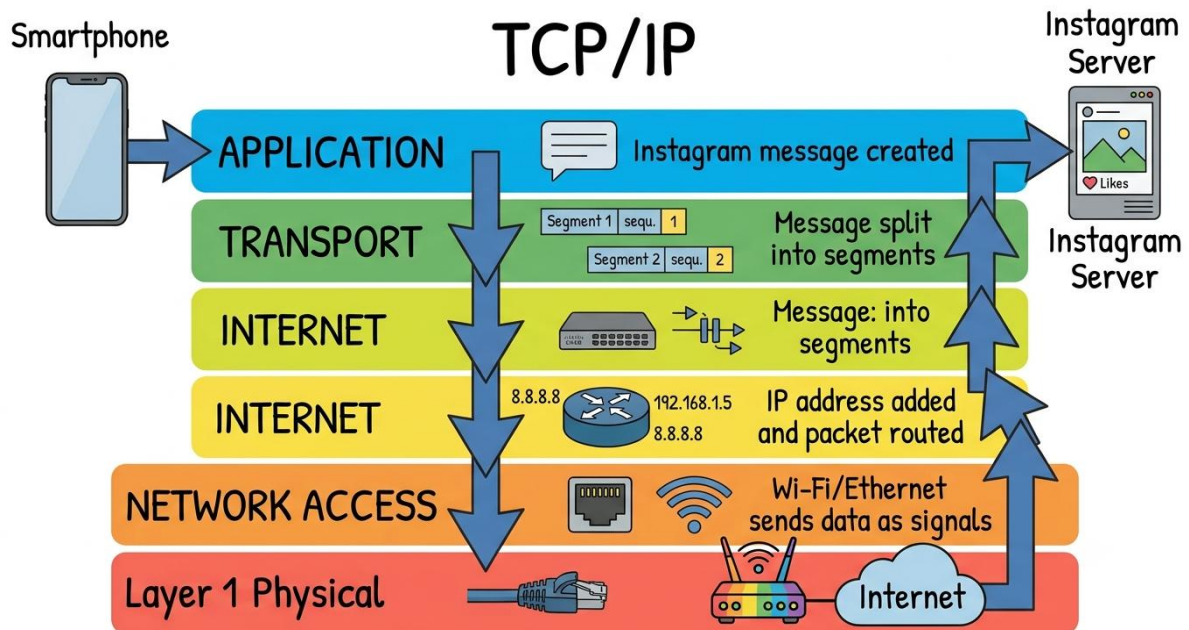
TCP Header Fields

Field	Purpose
Source Port	Identifies the port number of the device sending the data.
Destination Port	Identifies the port number of the receiving application.
Sequence Number	Helps arrange TCP data in the correct order.
Acknowledgment Number	Shows the next data number that the receiver expects.
Header Length	Shows the size of the TCP header.
Flags	Controls TCP operations such as connection setup and termination.
Window Size	Shows how much data the receiver can accept.
Checksum	Helps detect errors in the TCP segment.
Urgent Pointer	Provides information about urgent data when the URG flag is used.

Result:

The TCP header contains important information that helps TCP provide reliable communication between devices.

Task 4 — Device to Instagram Server



Explanation

1. Application Layer

I create and send a message using Instagram. The application prepares the data for communication.

2. Transport Layer

The message is divided into smaller pieces and transport information such as port numbers is added.

3. Internet Layer

IP addresses are used so that the data can travel from my device towards the Instagram server.

4. Network Access Layer

The data is sent through the local network using Wi-Fi or Ethernet and converted into signals for transmission.

Simple Flow

Instagram Message → Transport → IP → Wi-Fi/Ethernet → Internet → Instagram Server

Task 5 — Wireshark Packet Analysis

I selected one packet from my Wireshark capture for analysis.

Protocol: TCP

TCP/IP Layer: Transport Layer

How I identified it:

I identified the packet by checking the Protocol column in Wireshark. It showed **TCP**, and the packet details contained TCP information such as source port, destination port and sequence number.

Real-world action:

This type of packet can be part of the communication taking place when I open a website or use an application that uses TCP-based communication.

Result:

Wireshark helped me connect the packet information with the TCP/IP Transport layer and understand how network communication happens in real applications.